

Julius AI Workshop – Data Exploration Guide

This guide helps you explore the provided datasets using Julius AI.

It includes example questions (goals) and short explanations of key terms you might encounter during your analysis.

Subscription Code: PYLADIES

Possible Questions & Goals

Each question can be asked in Julius AI or used as inspiration to form your own research goal.

They're grouped by difficulty level to match different experience levels.

Basic – Get a Feeling for the Data

Customers

- How many customers are in the dataset?
- From which cities or countries do most customers come?
- What channels (email, social media, store) do customers prefer?
- Are there more new or returning customers?

Campaigns

- How many campaigns were run overall?
- Which campaign type was used most often?
- What is the typical campaign duration?

Products & Transactions

- Which products are the most common?
 - What day of the week seems to have the most sales?
 - Which payment method is used most often?
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Intermediate – Find Simple Relationships

Customers

- Do returning customers spend more than new customers?
- Are customers from certain regions more active?
- Does the preferred channel (e.g., email vs. social media) affect spending?

Campaigns

- Do campaigns with higher budgets perform better?
- Which campaign channels show the highest response or ROI?
- How does campaign duration influence success?

Products & Transactions

- Which product categories bring in the most revenue?
- Do discounts lead to higher total sales?
- Which months show the strongest sales trends?

Advanced – Explore Deeper Questions

Customers & Retention

- What factors are most related to high customer value (CLV)?
- Do customers who joined certain campaigns stay longer?
- How does customer loyalty differ by region or income group?

Campaign & ROI

- Which campaigns achieved the best ROI relative to their budget?
- Is there a link between campaign type and long-term customer value?

Products & Transactions

- Which products are often bought together?
- How do price changes affect total sales over time?
- Are certain products linked to high-value customers?

Key Terms & Concepts

These short definitions help you interpret and discuss your findings.

Customer Metrics

Term	Description	What it tells you
Customer Lifetime Value (CLV)	The total worth of a customer over their entire relationship with a company. It sums up all the revenue a customer generates across their lifetime.	Shows how valuable a customer is in the long run and helps prioritize high-value segments.
Retention Rate	The percentage of customers who stay engaged with a company or product over a specific time period.	Indicates customer loyalty and how well the company keeps people coming back.
Churn Rate	The percentage of customers who stop using a company's product or service during a given period.	Reveals customer loss — high churn may signal dissatisfaction or strong competition.

Campaign Metrics

Term	Description	What it tells you
ROI (Return on Investment)	Measures how profitable a campaign is by dividing the profit generated by its total cost. Formula: $(\text{Revenue} - \text{Cost}) / \text{Cost}$.	Helps evaluate if marketing spend is paying off or needs adjustment.
Conversion Rate	The percentage of people who complete a desired action (e.g., make a purchase, sign up, click a link). Calculated as: $\text{Conversions} / \text{Total Audience} \times 100$.	Shows how effective a campaign is at turning interest into action.

Data & Analysis Basics

Term	Description	What it tells you
Correlation	When two variables change together — either positively (both rise/fall) or negatively (one rises, the other falls).	Suggests a potential relationship, but not necessarily a cause. Useful for discovering patterns worth exploring further.
Segment	A defined group of similar customers or records based on shared traits (e.g., age group, location, spending level).	Helps compare behaviors and tailor strategies to specific audiences.
Trend	A consistent pattern or direction in data over time — such as increasing sales or declining engagement.	Shows long-term change and helps identify opportunities or risks.
Metric	A measurable value that tracks performance or progress toward a goal (e.g., ROI, CLV, conversion rate).	Provides objective indicators to evaluate how something performs.
Cleaning	The process of identifying and fixing wrong, missing, or inconsistent data values.	Ensures results are reliable and that insights reflect reality rather than data errors.
Outlier	A data point that is significantly different from the rest of the dataset.	May signal an error, but can also reveal an interesting or exceptional case.
Distribution	How data values are spread — for example, how many customers fall into each income range.	Helps understand data shape, detect skew, and spot unusual concentrations.
Normalization	Adjusting values to a common scale without distorting differences in ranges.	Makes it easier to compare variables with different units or magnitudes.
Aggregation	Summarizing detailed data into higher-level information (e.g., average sales per month).	Simplifies complex datasets and highlights broader trends.